

Protocol for *in vitro* transcription of DNA oligos by T7 polymerase

Template design:

T7 polymerase requires a double-stranded promoter region for efficient template binding, but can then transcribe a single-stranded DNA template. Therefore, the following protocol uses an approach, whereby two oligos are annealed in order to generate a dsDNA promoter, followed by a single-stranded overhang that contains the reverse complement of the desired RNA sequence.

- The reverse complement of the desired RNA sequence is ordered as a DNA oligo (assuming the length does not exceed ~80 nt), to which the reverse complement of the T7 promoter is appended at the 3' end
 - Example template:
 - 5'-TTTCTTAGCTGCCTATACGGCAGTGAA**CCTATAGTGAGTCGTATTA**-3'
 - **XXX** = reverse complement of T7 promoter
 - **XXX** = reverse complement of the desired RNA sequence
 - **Note:** the consensus T7 promoter sequence contains GGGAGA as the first 6 nucleotides that are transcribed. However, only the GG dinucleotide is required for high efficiency transcription (Ambion).
- A separate, synthetic DNA oligo is ordered that contains the T7 promoter in the forward direction, which can be annealed with the above construct:
 - T7oligo: 5'-TAATACGACTCACTATA-3'
 - **Note:** according to Ambion, for transcription of synthetic oligonucleotides, only the -17 to -1 positions of the promoter need to be double-stranded. This oligo takes this into account.

Stock solutions:

5x transcription buffer:	10 ml
150 mM Tris-Cl, pH 8.1	1.5 ml
125 mM MgCl ₂	1.25 ml
0.05% Triton X-100	50 µl
10 mM spermidine	100 µl
store at -20 °C	

10x TBE 1 L
108 g Tris base
55 g Boric acid
40 ml 0.5 M Na₂EDTA pH 8.0
H₂O to 1 L

0.5 M Na₂EDTA 0.5 L
93 g disodium ethylenediaminetetraacetate 2H₂O
10 g NaOH to get pH 8.0
H₂O to 500 ml
autoclave

10 % (w/v) APS 50 ml
5 g ammonium persulfate
H₂O to 50 ml
store at -20 °C

1 M spermidine
0.7 g spermidine
H₂O to 5 ml
store at -20 °C

5N NaOH 25 ml
5 g NaOH
H₂O to 25 ml

1N NaOH 20 ml
4 ml 5N NaOH
H₂O to 20 ml

10 % Triton X-100 10 ml
1 ml Triton X-100
9 ml H₂O
store at -20 °C

1% (w/v) bromphenol blue 10 ml
100 mg bromphenol blue
H₂O to 10 ml

1% (w/v) xylene cyanol 10 ml
100 mg xylene cyanol
H₂O to 10 ml

2x RNA Loading Dye	30 ml
95% formamide	28.5 ml
1 mM EDTA	60 µl of 0.5 M
0.025% SDS	75 µl of 10%
0.025% bromphenol blue	750 µl of 1%
0.01% xylene cyanol	300 µl of 1%
store at -20 °C	

Prepare 10 uM hybridized template:

1. Mix the following in a final volume of 100 µl:

20 µl 5x transcription buffer
1.5 nmol T7oligo (15 uM final)
1.0 nmol template (10 uM final)
DEPC H₂O to 100 µl

2. Heat at 70-80 °C, 2 minutes

3. Slow cool on bench-top
4. Use immediately or store at -20 °C.

Note: for some applications, prior gel purification of the DNA template may be desirable. However, for my transcriptions, the oligos are not very long and I immediately gel purify the RNAs after transcription, so purity of the DNA oligo is not a major concern.

Preparation of NTPs:

1. Add 500 µl of RNase-free H₂O to the 70-80 mg of each NTP.
2. Bring pH of each NTP to 6.8-7.2 by adding 5 µl of 5N NaOH followed by 10-90 µl of 1N NaOH. Monitor pH on the stripes with 6.5-10 range. Most adjustment will be needed for ATP, least – for UTP.
3. Measure absorbance of 500 fold dilution: 3 µl to 1.5 ml of H₂O.
 ATP λ_{max} 259 nm, ϵ_{max} $1.59 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$
 CTP λ_{max} 271 nm, ϵ_{max} $0.9 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$
 GTP λ_{max} 253 nm, ϵ_{max} $1.37 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$
 UTP λ_{max} 259 nm, ϵ_{max} $1.59 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$
4. Calculate concentrations and required dilutions to reach 100 mM concentration each.

Transcription reaction:

1. Mix the following in a 100 µl reaction:

20 µl 5X transcription buffer

5 µl 100 mM ATP

5 µl 100 mM CTP

5 µl 100 mM GTP

5 µl 100 mM UTP

1 µl 1M DTT

0.1 µl 1 mg/mL pyrophosphatase (Roche)

10 µl 1 mg/mL T7 polymerase

10 µl 10 uM hybridized template

DEPC H₂O to 100 µl

Final concentrations:

(1X)

(5 mM)

(5 mM)

(5 mM)

(5 mM)

(10 mM)

(1 ug/ml)

(100 ug/ml)

(1 uM)

Note: my typical final yields on transcription reactions of this size are roughly 30 µl of 10-20 uM (300-600 pmol). For my biochemical purposes, this is more than enough material. However, the above reaction can easily be scaled up.

2. Incubate at 37 °C, overnight.
3. DNase treat:

a) Add 5 units (5 µl) RNase-free DNase (Promega) directly to reaction

b) Incubate at 37 °C, 30 minutes

Note: this step is optional and may be omitted.

4. Add equal volume (100 µl) gel loading buffer directly to transcription reaction.

Note: my RNAs run very close to the xylene cyanol dye. Therefore, I use a gel loading buffer that contains only bromophenol blue.

5. Gel-purify transcribed RNAs on a medium thickness, denaturing urea-polyacrylamide gel.

Note: For my RNAs (~30 nt) I use a 15% gel. At this polyacrylamide concentration, with RNA from a 100 µl transcription reaction, I get excellent separation of the desired RNA length from n+1 and n-1 contaminants (these arise from template-independent addition of nts by T7 polymerase; and synthetic template oligos that are not the full length). Optimization of the gel running time and acrylamide concentration will be necessary for gel-purifying RNAs of different length.

Gel-purification:

PAGE Fragment Resolution: Denaturing Conditions (6M Urea)

% Acrylamide	Fragment Size	Bromophenol Blue	Xylene Cyanol
30	2-8	6	20
20	8-25	8	28
10	25-35	12	55
8	35-45	19	75
6	45-70	26	105
5	70-300	35	130
4	100-500	~50	~230

1. Clean glass plates with EtOH and coat with 200 µl Silanization Solution I.
2. Assemble glass plates.
3. Prepare 15% urea polyacrylamide gel.
4. Denaturing Urea Polyacrylamide Gel (300 ml for a big gel: 35 × 25 cm)

	10%	12.5%	15%	
10x TBE	30	30	30	ml
40% Acrylamide-bisacrylamide (19:1)	75	93.75	112.5	ml
Urea	108	108	108	g
Water	192	173	155	ml
Before casting add:				
10 % APS	1.5	1.5	1.5	ml
TEMED	600	600	600	µl

5. After addition of APS and TEMED to Urea-Polyacrylamide mixture pour the gel immediately.
6. Place comb and polymerize for 60 min.
7. Assemble electrophoresis apparatus.
8. Remove comb.
9. Fill inner and outer buffer chambers with 1x TBE.

10. Rinse wells with 1x TBE buffer.
11. Pre-run the gel for 30 min to heat-up and remove remaining urea. Set constant P=40 W, 3000 V, 400 mA
12. Load sample.
13. Run P=40 W, 3000 V, 400 mA for around 3 h, till the bromphenol blue band is migrating in the last $\frac{1}{4}$ of the gel.
14. Remove one of the glass plates and cover the gel with saran wrap. Flip. Remove another glass plate and cover all gel with saran wrap.
15. UV shadow band and excise with sterile razor and place in Falcon tube.
16. Break gel slices with p-1000 tip.
17. Crush and soak gel slices in equal volume of RNase-free water.
18. Elute RNA overnight by rocking at 4 °C.

Next day:

19. Filter out gel pieces by spinning sample through Corning Spin-X filter tube. Centrifuge at top speed for 5 minutes, 4 °C.

Note: I cut the top off a P-1000 tip so that I can pipette as much as possible of the water-gel slurry into the Corning filter tube.

20. Add 30 μ l 10x NEBuffer 3 to each sample, then 1 μ l CIP (10 U/ μ l; NEB). Incubate at 37 °C for one hour.

Note: this step is only required if removal of the 5' triphosphate is required (e.g. if the RNA will subsequently be 5'-radiolabeled with T4 PNK.)

21. Phenol/chloroform extract with 300-500 μ l phenol-chloroform solution, pH 8.0.
22. Ethanol precipitate the RNA, by adding 1/10 volume 3 M NaOAc (pH 5.2), 1/100 volume 100x linearized acrylamide and 3 volumes 100% ethanol.
 - a) Incubate at -80 °C, at least 1 hour
 - b) Spin down at top speed, 20 minutes, 4 °C. Remove supernatant.
 - c) Wash pellet with 70% ethanol. Spin down again and remove supernatant.
 - d) Dry pellets in speed vacuum, ~10 minutes.

14. Resuspend pellets in 30 μ l DEPC H₂O (or buffer) and store at -20 °C.